

WEB APPLICATION SECURITY BASICS REPORT

Cybersecurity Internship Task

Prepared By: Gudla Sai Abhishith

Tools Used:

- **Burp Suite**
- **Altoro Mutual Demo Application**
- **Web Browser**

1. Objective

The objective of this task was to identify basic web application vulnerabilities in a safe demo environment using Burp Suite and manual testing techniques.

2. Cross-Site Scripting (XSS) Testing

A reflected Cross-Site Scripting vulnerability was tested on the search functionality of the demo application. A JavaScript payload was injected into the search parameter and executed successfully in the browser.

3. SQL Injection (SQLi) Testing

A SQL Injection vulnerability was identified in the login functionality. The SQL comment operator was used to bypass password verification and gain unauthorized access to the admin account.

4. Findings

The testing demonstrated improper input validation, reflected XSS vulnerability, and SQL Injection authentication bypass vulnerabilities within the application.

5. Mitigation Recommendations

Use parameterized queries, implement proper input validation and output encoding, sanitize user input, and apply Content Security Policy headers.

Tool	Purpose
Burp Suite	Intercepting and modifying HTTP requests
Altoro Mutual	Safe demo application for testing
Browser	Testing web application functionality

Vulnerability	Impact
Reflected XSS	JavaScript execution in browser
SQL Injection	Authentication bypass
Improper Input Validation	Unsafe processing of user input

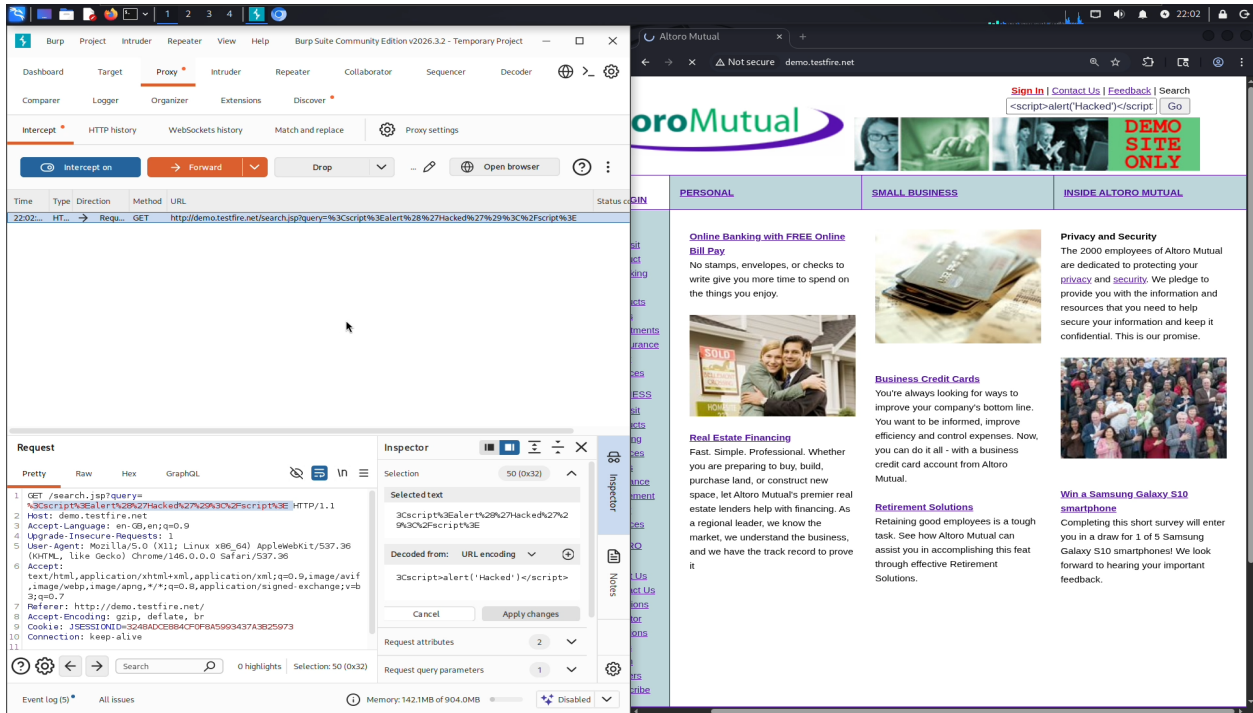


Figure 1: Burp Suite intercepting XSS payload

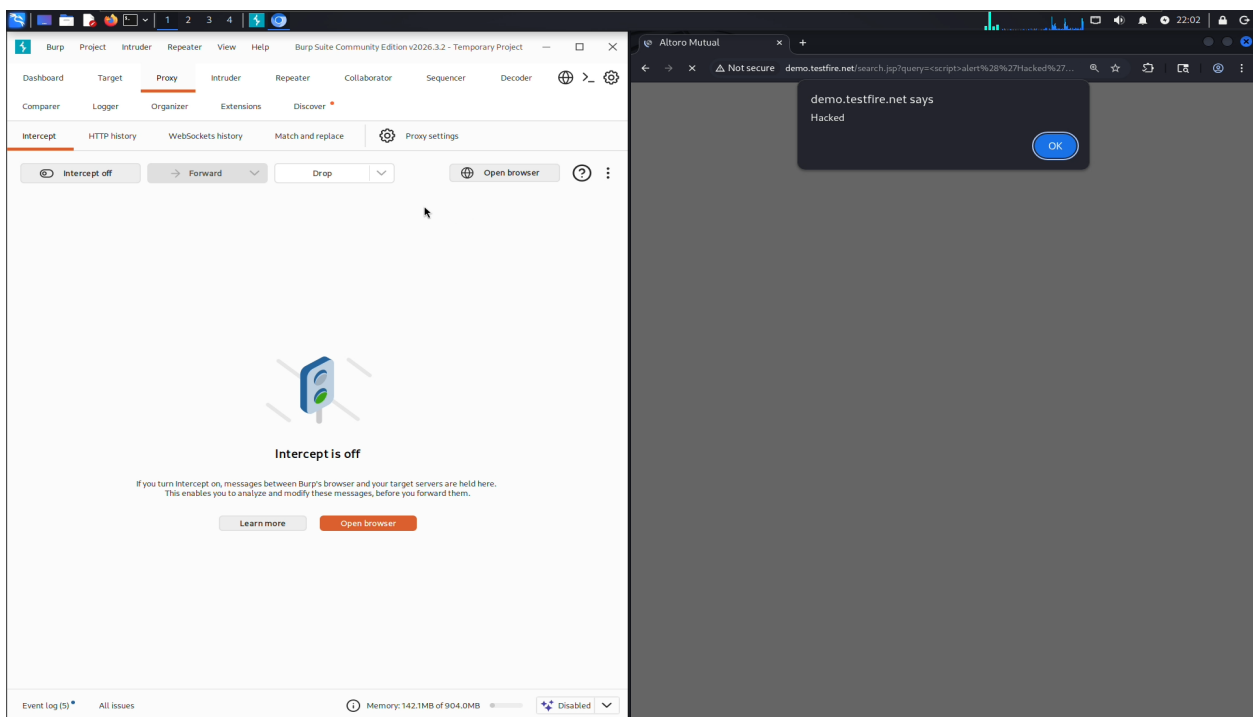


Figure 2: Successful XSS alert execution

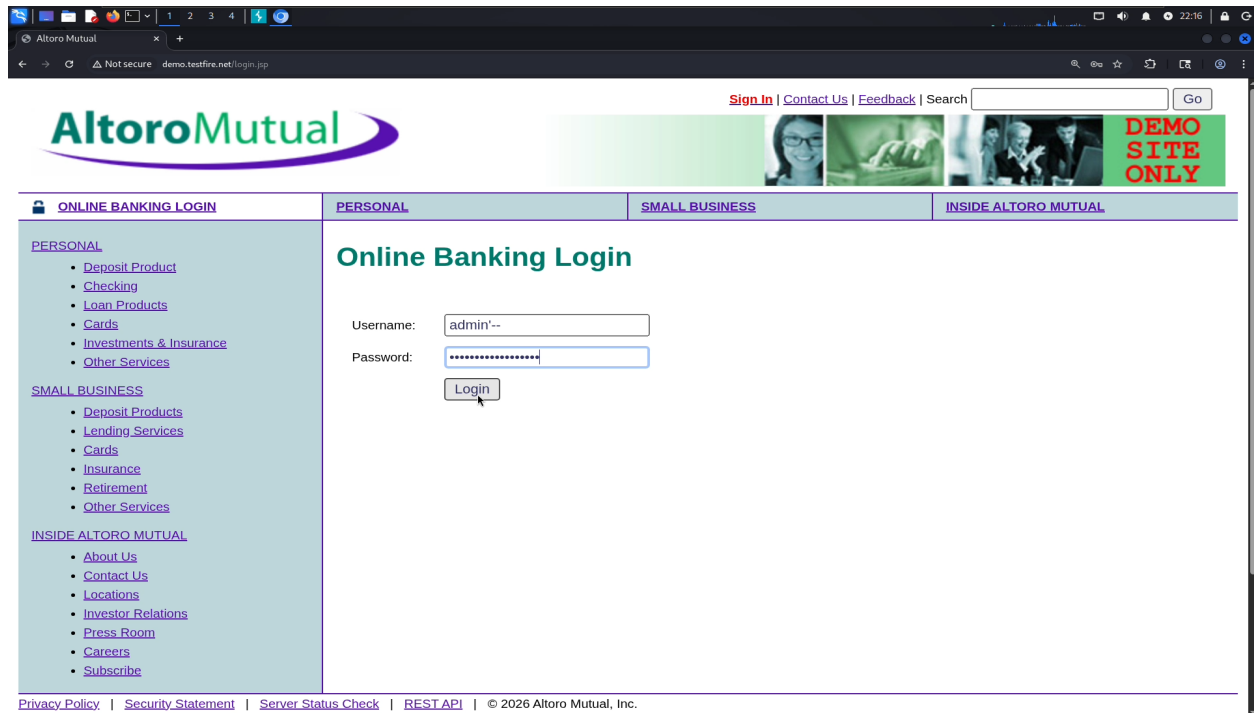


Figure 3: SQL Injection payload in login form

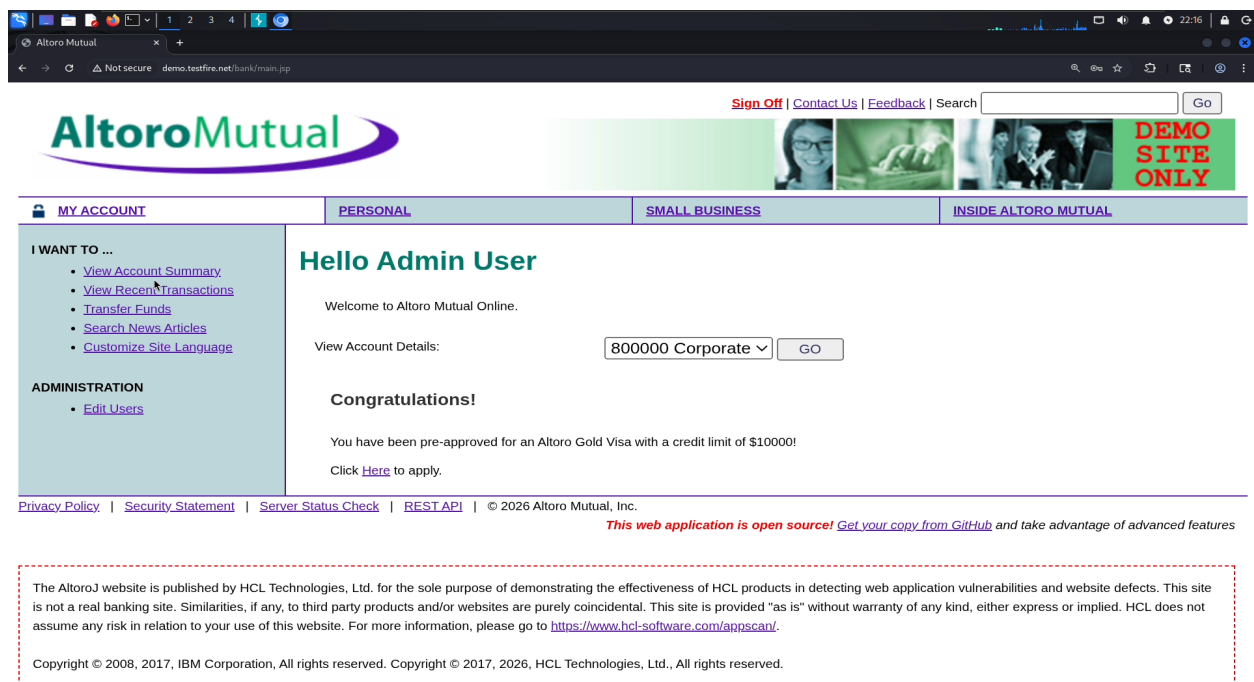


Figure 4: Successful admin login bypass

6. Conclusion

This task provided practical experience in identifying common web application vulnerabilities using Burp Suite and safe demo applications. The assessment demonstrated how insecure input handling can lead to serious security risks such as XSS and SQL Injection.